

Functions and transformations - Set 2 - Solutions

Specification MM1.7, MM8.2, MM8.3, MM8.4.

Answer key

Question	Answer	Topic
1	D	Functions and transformations
2	A	Functions and transformations
3	F	Functions and transformations
4	F	Functions and transformations
5	D	Functions and transformations
6	F	Functions and transformations

Worked solutions

Question 1 FUN-04 · Functions and transformations · spec MM1.7

Which of the following functions, each with domain the set of all real numbers, is one-to-one?

Answer: D

$x^3 + x$ is strictly increasing (its derivative $3x^2 + 1 > 0$), so different inputs give different outputs. $x^3 - x$ takes the value 0 at $x = -1, 0, 1$; the others are even or periodic.

Full method - how and why

'One-to-one' means no horizontal line meets the graph twice. The quickest test is whether the function is always increasing (or always decreasing).

- 1 Rule out the even and periodic functions: $|x|$ and $x^2 + 2x$ are symmetric (e.g. $|1| = |-1|$), and $\cos x$ repeats every 2π .

why: Any symmetry or repetition immediately produces two inputs with the same output.

- 2 Test $x^3 - x$: it equals 0 at $x = -1, 0, 1$ - three inputs with the same output.

why: A cubic with three distinct roots is certainly not one-to-one.

- 3 Test $x^3 + x$: its derivative $3x^2 + 1$ is always positive, so the function is strictly increasing.

why: A strictly increasing function can never take the same value twice, which is exactly one-to-one.

- 4 So $f(x) = x^3 + x$ is the one-to-one function.

why: Alternatively, x^3 and x both increase, so their sum does.

Where it goes wrong

- Assuming every cubic is one-to-one - $x^3 - x$ is a counterexample.
- Thinking $|x|$ is one-to-one because it has a 'simple' formula.

Takeaway One-to-one $\Leftrightarrow$ strictly monotonic; check for symmetry or repeated values first, then look at the derivative's sign.

Question 2 FUN-07 · Functions and transformations · spec MM8.2

The function f satisfies $f(x) = x^2$ for all x . Which of the following transformations maps $y = f(x)$ onto $y = 4x^2 - 8x + 4$?

Answer: A

$4x^2 - 8x + 4 = 4(x - 1)^2$, i.e. $y = 4f(x - 1)$: translate 1 right, then stretch vertically by 4. (Equivalently $y = f(2x - 2) = f(2(x - 1))$: stretch parallel to the x -axis by factor $\frac{1}{2}$ then translate 1 right - not offered.)

Full method - how and why

Work backwards: put the target into a form that shows the transformations. Completing the square reveals them.

- 1 Factorise the target: $4x^2 - 8x + 4 = 4(x^2 - 2x + 1) = 4(x - 1)^2$.

why: Taking out the 4 exposes a perfect square - the shape of f with a translation.

- 2 Read it as $y = 4f(x - 1)$ where $f(x) = x^2$.

why: The inside $(x - 1)$ is a translation, the outside 4 is a vertical stretch.

- 3 Translate 1 unit in the positive x -direction, then stretch parallel to the y -axis by factor 4.

why: With one inside and one outside change, the order between them does not matter - but the direction of each does.

- 4 Confirm with a point: the vertex $(0, 0)$ maps to $(1, 0)$, and the target $4(x - 1)^2$ does have its vertex at $(1, 0)$ ✓.

why: A point check catches direction errors instantly.

Where it goes wrong

- Translating left $(x + 1)$, giving $4(x + 1)^2 = 4x^2 + 8x + 4$ - the wrong sign on the middle term.
- Reading the 4 as a horizontal stretch; $y = f(2x) = 4x^2$ has no translation at all.

Takeaway Factor the target into $af(x - h) + k$ form; the numbers then name the transformations directly.

Question 3 FUN-10 · Functions and transformations · spec MM8.2

$f(x) = x^2 + 1$. Which of the following statements is/are true?

- I $f(f(x)) = x^4 + 2x^2 + 2$ for all x
- II $f(x + 1) = f(x) + 1$ for all x
- III $f(2x) = 4f(x) - 3$ for all x

Answer: F

I: $f(f(x)) = (x^2 + 1)^2 + 1 = x^4 + 2x^2 + 2$; true. II: $f(x + 1) = x^2 + 2x + 2 \neq x^2 + 2$; false. III: $f(2x) = 4x^2 + 1$ and $4f(x) - 3 = 4x^2 + 1$; true.

Full method - how and why

Three algebraic claims about one function. Each is settled by direct substitution - and the false one needs only a single value of x .

- 1 I: $f(f(x)) = (x^2 + 1)^2 + 1 = x^4 + 2x^2 + 1 + 1 = x^4 + 2x^2 + 2$. **True.**
why: Substitute the whole of $f(x)$ into f ; expand $(x^2 + 1)^2$ carefully - the cross term $2x^2$ is essential.
- 2 II: $f(x + 1) = (x + 1)^2 + 1 = x^2 + 2x + 2$, whereas $f(x) + 1 = x^2 + 2$. These differ by $2x$. **False.**
why: A single value settles it: $x = 1$ gives $f(2) = 5$ but $f(1) + 1 = 3$.
- 3 III: $f(2x) = 4x^2 + 1$ and $4f(x) - 3 = 4x^2 + 4 - 3 = 4x^2 + 1$. Equal for all x . **True.**
why: Compute both sides independently and compare - do not try to 'see' it.
- 4 I and III only.
why: Map the verdicts onto the option list.

Where it goes wrong

- Assuming II is true because functions 'usually' behave additively - almost none do.
- Expanding $(x^2 + 1)^2$ as $x^4 + 1$.

Takeaway Test each claim by substitution; one numerical counterexample is enough to destroy a 'for all x ' statement.

Question 4 FUN-12 · Functions and transformations · spec MM1.7

Let $f(x) = \sqrt{x-3}$ for $x \geq 3$ and $g(x) = x^2 + 3$. Which of the following statements is/are true?

- I $g(f(x)) = x$ for all $x \geq 3$
- II $f(g(x)) = x$ for all real x
- III $f(g(x)) = |x|$ for all real x

Answer: F

I: $g(f(x)) = (\sqrt{x-3})^2 + 3 = x$ for $x \geq 3$: true. $f(g(x)) = \sqrt{x^2 + 3} = |x|$, so II fails (e.g. $x = -1$) and III is true.

Full method - how and why

Two functions that almost invert each other. The asymmetry comes from $\sqrt{x^2} = |x|$ and from the restricted domain.

1 I: $g(f(x)) = (\sqrt{x-3})^2 + 3 = (x-3) + 3 = x$ for $x \geq 3$. **True.**

why: Squaring a square root recovers the inside, provided the inside is non-negative - guaranteed by the stated domain $x \geq 3$.

2 Compute the other order: $f(g(x)) = \sqrt{(x^2 + 3) - 3} = \sqrt{x^2} = |x|$.

why: Do the algebra once and read both remaining statements off it.

3 II claims this equals x for all real x : false at $x = -1$, where $|x| = 1 \neq -1$. **False.**

why: The square root returns the non-negative value, so it cannot reproduce a negative input.

4 III says it equals $|x|$, which is exactly what we found. **True.**

why: Same computation, correct conclusion.

5 I and III only.

why: Match to the options.

Where it goes wrong

- Assuming $f(g(x)) = x$ because f and g 'look like' inverses - inverses require matching domains and ranges.
- Writing $\sqrt{x^2} = x$.

Takeaway $f(g(x)) = x$ and $g(f(x)) = x$ are different statements; $\sqrt{x^2} = |x|$ is what breaks the symmetry here.

Question 5 FUN-16 · Functions and transformations · spec MM1.7

The function f is defined by $f(x) = |x - 2| + |x + 2|$ for all real x . What is the minimum value of $f(x)$, and for how many values of x is it attained?

Answer: D

For $-2 \leq x \leq 2$: $f(x) = (2 - x) + (x + 2) = 4$. Outside this interval $f(x) > 4$. So the minimum 4 is attained on the whole interval $[-2, 2]$.

Full method - how and why

A sum of two moduli. Splitting the line at the two 'corner' points shows the function is constant in the middle.

1 The moduli change behaviour at $x = 2$ and $x = -2$; consider the three regions separately.

why: $|x - a|$ changes formula at $x = a$; the critical points split the line into pieces where the function is linear.

2 For $-2 \leq x \leq 2$: $|x - 2| = 2 - x$ and $|x + 2| = x + 2$, so $f(x) = (2 - x) + (x + 2) = 4$.

why: The x terms cancel exactly - the function is **constant** on this whole interval.

3 For $x > 2$: $f(x) = (x - 2) + (x + 2) = 2x > 4$. For $x < -2$: $f(x) = (2 - x) + (-x - 2) = -2x > 4$.

why: Outside the middle the function grows, so nothing beats 4.

4 Minimum 4, attained for every x in $[-2, 2]$: infinitely many values.

why: The second half of the question is the real test - most students find the 4 and then guess.

Where it goes wrong

- Answering 'attained for exactly one x ' by assuming a minimum must occur at a single point.
- Testing only $x = 0$ and concluding the minimum is 4 without checking whether it is attained elsewhere.

Takeaway $|x - a| + |x - b|$ is constant (and minimal) between a and b - it measures total distance to two fixed points.

Question 6 X2-11 · Functions and transformations · spec MM8.2

Let $f(x) = x^2$ for all real x , and let g be the function whose graph is obtained by translating the graph of $y = f(x)$ by 2 units in the positive x -direction and then reflecting in the y -axis. Which of the following statements is/are true?

- I** $g(x) = (x + 2)^2$
- II** $g(x) = (x - 2)^2$
- III** The graph of g has its vertex at $(-2, 0)$

Answer: F

Translation gives $(x - 2)^2$; reflecting in the y -axis replaces x by $-x$: $(-x - 2)^2 = (x + 2)^2$, with vertex at $(-2, 0)$.

Full method - how and why

Composite transformations, applied in the stated order. Each transformation acts on the function you currently have.

- 1 Translate $y = x^2$ by 2 in the positive x -direction: replace x by $x - 2$, giving $y = (x - 2)^2$.
why: A shift **right** subtracts inside the bracket - counter-intuitive but standard.
- 2 Now reflect in the y -axis: replace x by $-x$, giving $y = (-x - 2)^2$.
why: Reflection in the y -axis is the substitution $x \rightarrow -x$.
- 3 Simplify: $(-x - 2)^2 = (-(x + 2))^2 = (x + 2)^2$. So $g(x) = (x + 2)^2$ - **I is true, II is false.**
why: Squaring removes the overall minus sign.
- 4 III: $(x + 2)^2$ has its vertex where $x + 2 = 0$, i.e. at $(-2, 0)$. **True.**
why: Consistent: the vertex moved right to $(2, 0)$, then reflected to $(-2, 0)$.
- 5 So I and III.
why: Match to the list.

Where it goes wrong

- Doing the transformations in the wrong order - reflecting first then translating gives $(x - 2)^2$ instead.
- Shifting right by adding 2 inside the bracket.
- Forgetting to simplify $(-x - 2)^2$, and so failing to recognise it as $(x + 2)^2$.

Takeaway Right by a : $x \rightarrow x - a$. Reflect in the y -axis: $x \rightarrow -x$. Apply them in the order given, tracking the vertex as a check.